

Instructions for Exercise 5

1. This exercise is graded passed/failed. The early-bird deadline is **30.3.2025**. The final deadline is **16.4.2025**.
To pass the course, the student must complete 6 weekly exercises. However, if the student returns each exercise within the early bird date, 5 exercises are sufficient to pass the course.
2. Return 2 or 3 pdf-files
 - i. A pdf-file with the written exercise (1 page) and answers to the questions in the Python files
 - ii. A pdf-file with the Python Exercises 5.1 (credit risk) and 5.2 (Twitter sentiment analysis).
 - iii. **For the 5 and 6 credit version of the course:** A pdf-file with the Python Exercise 5.3 (macroeconomic indicators and the S&P 500 index).

A. Written exercise (Individual work)

Write a total of one page (400-500 words) on the following 2/3 topics. Don't write about things that are already trivial to you. The aim of this exercise is to learn new things.

1. *Credit scoring.* Credit scoring models must accurately forecast risk as their primary objective. Furthermore, regulators mandate that these models must be transparent and open to auditability.
 - a. What potential biases can machine learning models introduce in credit scoring and how can they affect lending decisions?
 - b. Reflect on the transparency of decisions. (why, challenges, possibility to cope)
 - c. How do changes in economic conditions and market dynamics affect the performance of ML models in credit scoring? (e.g., non-linear relationship, unprecedented events, good times/bad times)
2. *Textual data and sentiment data in trading.*
 - A. What kind of textual data and sentiment data can be used in trading?
 - B. The risks associated with using textual data/sentiment data can be divided into
 - a. data-driven issues
 - b. analysis and decision-making issues.

Give examples of both types of risk (short description). Try to find yourself some risks. You will learn nothing by asking ChatGPT.

When you have done this exercise, see what ChatGPT suggests. Based on this, fill in your answer in italics.

3. **5/6 credit version of the course:** *Macroeconomic indicators*. In the Python Exercise 5.3, you will use macroeconomic indicators to predict the S&P500 index returns. What macroeconomic indicators could be useful and why? Once you have done some independent thinking, you can look at the indicators used in exercise 5.3. Why were they chosen? It would be important for you to understand the financial market-relevant information contained in the macroeconomic indicators.

B. Python exercises

- Exercise 5.1 is about credit scoring. Read the code, answer the questions, and complete simple pieces of code.
- In Exercise 5.2, we apply Twitter feeds, sentiment analysis and deep learning to trading. Read the code, answer the questions, and complete simple pieces of code. **IMPORTANT:** We will consider NLP in Lecture 6 on 26.3. If you are completely new to NLP, you can look forward to this lecture. After that, you still have a few days until 30 March for an early bird dl.
- **5/6 credit version of the course:** go through Exercise 5.3 on macroeconomic indicators, S&P500 index and PCA.